Avi Gobrin

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Education

University of Toronto — 2023 – Apr 2027
Honours BSc, Mathematics & Its Applications Specialist, Statistics Major, Computer Science Minor — *Toronto, ON*

Relevant Coursework: Probability & Statistics I/II, Multivariate Data Analysis, Nonlinear Optimization, Real Analysis

Experience

Nextdoor — May 2026 – Aug 2026
Machine Learning Intern — *San Francisco, CA*

- Increased growth and weekly active users by launching Best Places for Families, an SEO ranking product covering ∼300,000 US neighbourhoods, building every layer end to end: internal data and Census/ACS ingestion through Airflow pipelines to a Postgres/GraphQL/React serving path.
- Designed the scoring methodology comparing 4 models predicting family-friendliness for the 197,556 of ∼300,000 neighbourhoods with sufficient data, screening 33 candidate metrics down to 8 features.
- Validated ranking quality through an evaluation harness before launch, testing rankings against a preregistered adversarial test set of neighbourhoods with pairwise Bradley-Terry judgments, cross-checked by both LLM and manual review.
- Made the system self-sustaining with scheduled jobs, tests, and monitoring, and built Databricks tooling that lets non-technical stakeholders retune feature weights and see how any city ranks.

Shopify — May – Aug 2025
Software Engineering Intern — *Toronto, ON*

- Increased merchant activation and retention by building a full-stack partner-attach system (Ruby on Rails, Calendly API) that screens each new merchant and books qualifying ones into guided onboarding with a vetted advisor. Routing over 100 merchants per day into the partner attach flow.
- Brought Shopify's entire German merchant base into VAT compliance in the 1.5 months from regulatory notice to launch. Reconciled what data Shopify already held against what merchants had to supply, integrated a difficult government tax API, and built a Rails intake form with an email flow that returns each merchant's compliance PDF.
- Cleared reliability-impacting technical debt across adjacent teams and codebases, increasing engineering output and velocity.

Research

University of Toronto, Department of Statistical Sciences — Expected: Fall 2026
Incoming Undergraduate Researcher, Advisor: Prof. Ricardo Baptista (Vector Institute) — *Toronto, ON*

- Investigating a unifying rank-reduction framework linking in-context learning and fine-tuning as low-rank weight updates, connecting recent transformer interpretability work to classical multivariate statistics.

Technical Reports

- Engagement, Performance, and Student Outcomes at the Open University (n = 23,301) [Report]
- Discovering Latent Structure in the IPIP Big Five Personality Inventory (n = 19,718) [Report]
- The Effect of Transaction Costs on Optimal Portfolio Rebalancing Frequency (Monte Carlo over 10,000 paths) [Report]
- What Makes a Wine Good? Chemical Structure and Quality in Portuguese Wines (n = 6,497) [Report]

Projects

Latent-Structure Modeling of Student Outcomes (OULAD) *Research Project* [Report] — 2026

- Modelled latent engagement and performance structure across 23,301 complete student-course registrations (of 25,820) via a parallel-analysis-validated 3-factor varimax model (Engagement $h^2 = 0.84$, Consistent Performance $h^2 = 0.72$) and 3-component PCA (72.9% variance); showed FA, GMM, and LDA converge on the same two latent dimensions.
- Benchmarked classifiers under 5-fold CV (logistic regression 67.0% vs. LDA 64.5% vs. 53.0% majority baseline) and showed LR lifts minority-class recall from 3.2% to 39.3%, diagnosing LDA's shared-covariance violation.

GPT from Scratch *Personal Project (in progress)* [Repo] — 2026

- Building a character-level GPT in pure NumPy, growing one codebase from a bigram baseline into a full Transformer (multi-head causal self-attention, LayerNorm, Adam, temperature/top-k sampling); every forward and backward pass is derived on paper before being coded, with no autograd; bigram and neural-bigram models complete.

Technical Skills

ML & Data: PyTorch, scikit-learn, NumPy, Pandas, SciPy, statsmodels, Matplotlib, Seaborn; neural nets and backprop implemented from scratch
Languages: Python, R, SQL, Java, JavaScript, Bash, HTML, CSS
Systems & Web: Airflow, Databricks, Postgres, Linux/Unix, REST APIs, GraphQL, React, Ruby on Rails, Node.js, Flask
Tools: Git, GitHub, Jupyter, LaTeX, Claude Code, Cursor, Claude, LLM agents, Maven, Excel